

Introduction to Artificial Intelligence

Course Description

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Why?

- **AI is Everywhere:** AI is a great tool, and knowing how it works improve your interaction with it.
- **Employability:** It is good on your CV, and many companies are looking for AI skills.
- **Interdisciplinary Impact:** AI techniques are applicable across various fields, from healthcare to finance.

Key Takeaways from This Course

By the end of this course, you will

- Learn foundational concepts and in AI
- Develop problem-solving skills for real-world challenges
- Understand the limits of certain AI techniques

Note: this course is only an introduction to AI. We cannot and will not cover all the approaches to AI, and we will not enter into details. For this, there are other specialized courses.

Course codes:

- AI501 – bachelor in CS – 7.5 ECTS
- AI501 – bachelor in AI – 7.5 ECTS
- DS843 – master in DS – 7.5 ECTS + $\underbrace{2.5 \text{ ECTS}}_{\text{project}}$

Teacher:

- Matteo Acclavio (acclavio@imada.sdu.dk)

TAs:

- Andrea Blasi Núñez
- William Alexander Lydum Senning
- Dimitra Vasilakou (DS843 only)

Exam

Written exam for all

- Structure:
 - Simple multiple choice questions (MCQ): covering broadly all the notions of the course
 - Open questions (OQ): require a bit of reasoning
- Grade
 - if $> 80\%$ on MCQ, then the grade is a combination of the weight of both parts ($\text{MCQ} \times 0.3 + \text{OQ} \times 0.7$)
 - if $> 60\%$ on MCQ and pass on OQ, then max grade 2
 - if $< 60\%$ on MCQ or not pass on OQ, failed

Only for DS843

- project [next slide]

Project for DS843

Group project with written report

- Up to 4 people
- Possibility to have personalized projects. For example:
 - Try to “solve” a game you love
 - Pick a chapter in the book introducing concepts/ideas useful to address a concrete problem you have and try to apply them
 - Propose your own idea
- If no ideas I will provide a default project (~ Eastern)
- Grade can go up or down by 1 grade over written exam [both written exam and project must be done to pass]

How to prepare for the exam?

Teaching Modality

- Traditional frontal lectures (2h a week)
- Teaching assistant (TA) sessions for exercises (2h a week)

Textbook

"Artificial Intelligence: A Modern Approach" (4th Global Edition)

[Free PDF can be found online. The library has physical copies.]

- No need to read the book before classes: attend the lecture, and then read it at home to consolidate the notions.
- No need to do exercises before the class: attend TA sessions, do exercises during the TA sessions, check and correct the exercises with the TAs during the TA sessions.
- Slides will be made available after each lecture. However...

How to reach me?

- After lectures, during breaks
- By email: acclavio@imada.sdu.dk
- Drop by my office (Ø16-601a-1)
- Remember there are also the TAs available during TA sessions!

Tentative Plan

Weeks 6-7: Introduction to AI and Intelligent Agents

- History (Chapter 1)
- Intelligent Agents (Chapter 2)
- Limits and Ethics (Chapter 28)

Weeks 8-11: Search

- Simple search (Chapter 3)
- Search in complex environments (Chapter 4)
- Constraint Satisfaction Problems (Chapter 5)
- Adversarial Games (Chapter 6)

Weeks 12-13, 15: Logic [Given by Dr. Giulia Manara]

- Propositional logic (Chapter 7)
- First-order logic (Chapter 8)
- Automated reasoning (Chapter 9)

Weeks 16-18: Probabilistic reasoning

- Notions on Probability (Chapter 12)
- Bayesian Networks (Chapter 13)

Weeks 19-20: Advance Topics + tools

Let's start!